

Subject Title (Theory): Software Engineering

Course code	Type of Course	Theory / Practical	Credits	Instruction hour per week	Duration of Exam	Formative Assessment Marks	Summative Assessment Marks	Total Marks
E5BCA2T1	Specialization-1	Theory	04	04	3hrs	20	80	100

Course Outcomes (COs): At the end of the course, students will be able to:

CO1: How to apply the software engineering lifecycle by demonstrating competence in communication, planning, analysis, design, construction, and deployment.

CO2: An ability to work in one or more significant application domains.

CO3: Work as an individual and as part of a multidisciplinary team to develop and deliver quality software.

CO4: Demonstrate an understanding of and apply current theories models, and techniques that provide a basis for the software lifecycle.

CO5: Demonstrate an ability to use the techniques and tools necessary for engineering practice.

Unit I:

15 hrs.

The Product and The Process: Evolving role of software, software characteristics and components, Crisis, Software Myths, Software Engineering-A Layered Technology, Software process, linear sequential model, Prototyping model, RAD model, Evolutionary software process model.

Project Management Concepts: The Management Spectrum, The People, The Product, The Process, and The Project. W5HH Principle.

Software Process and Project Metrics: Measures, Metric Indicators, Metric in process and the Project Domains, Software Measurement, Metrics for software quality.

Unit II:

15 hrs.

Software Project Planning: Project Planning Objectives, Software Project Estimation, decomposition Techniques, Empirical Estimation Models.

Risk Analysis and Management: Software Risks, Risk Identification, Risk Projection, Risk Refinement and Risk Mitigation, Monitoring, and Management.

Unit III:

15 hrs.

Analysis Concepts and Principles: Requirement analysis, communication techniques, analysis principles, software prototyping and specification.

Analysis Modeling: Elements of analysis model, data modeling, functional modeling, behavioral modeling, the mechanics of structured analysis, data dictionary, other classical analysis methods.

Unit IV:

15 hrs.

Design Concepts and Principles: Software design and software engineering design process, design principles, design concepts, design methods, data design, architectural design and process, transform and transaction mappings, design post processing, architectural design optimization, interface design, procedural design.

Software Testing Techniques and Strategies: Fundamentals, Test case design, White box testing, Basis path testing, Control structure testing, Black box testing, Software testing strategies.

Text Books:

1. Software Engineering, Fifth Edition, Roger - Pressman, McGraw Hill.
2. Software Engineering, Ian Sommerville, International Computer Science, Series

References:

1. Software Engineering, Schooma, McGraw Hill
2. Object Oriented Design and Analysis, Booch, Benjamin / Cummings,
3. Software Engineering: A Practitioner's Approach 7th Edition, Roger – Pressman, Tata McGraw - Hill Education (2010)

Formative Assessment for Theory	
Assessment Type	Marks
Internal Assessment Test 1	05
Internal Assessment Test 2	05
Assignment	10
Total	20